Does Market-Capitalization Rank Predict Returns?

A falsification test of a rules-based mega-cap strategy, 1995–2025

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Abstract

This note tests whether a concentrated portfolio of the ten largest S&P 500 companies by market capitalization, held with annual rebalancing and systematic sell rules, outperforms a passive index position. The motivating hypothesis is that changes in market-capitalization rank carry forward-looking information: companies entering the top ten arrive on sustained growth, and companies existing have begun a structural decline.

The hypothesis is not supported, and the finding holds across both windows tested. Over 1995–2025 the strategy returned 11.03% compounded against 11.14% for the S&P 500 Total Return index, effectively a tie-on raw return, but achieved at 22.4% volatility against 18.1%, giving a Sharpe ratio of 0.49 against 0.57. Over 2000–2025 it returned 5.98% against 8.11%, a gap of 2.13 percentage points a year, losing on every measure.

The 1995–2025 window is the more instructive of the two. The strategy produced a higher arithmetic mean annual return than the index, 13.39% against 12.76% and still finished behind on compound return. Section 5 shows that the entire 0.63-point advantage was consumed by volatility drag, and that this is arithmetically exact rather than approximate.

1. Hypothesis

Market-capitalization rank is backward-looking, but changes in rank may not be. A company entering the top ten has typically done so through sustained revenue and earnings growth; a company exiting has typically begun a structural decline. If either effect persists beyond the point at which rank changes, a portfolio holding the current top ten and systematically removing deteriorating constituents should outperform a passive index.

The test is deliberately mechanical. No view is taken on any individual business and no discretion is applied, so what is being tested is whether the rank signal alone carries information, not whether a human reading the same list could pick well from it.

2. Data and Methodology

Rankings are the twenty largest S&P 500 companies by market capitalization at each year-end, from FinHacker, shifted forward one year so that each portfolio forms on the January ranking a contemporaneous investor would have seen. These are point-in-time rankings, not reconstructed from present-day knowledge of which companies grew large; the backtest is therefore free of look-ahead bias in constituent selection. Rankings are available from 1990, and the windows tested begin in 1995 and 2000.

Prices are daily closes adjusted for splits and dividends, from Yahoo Finance. Because holdings therefore earn total return, the benchmark is the S&P 500 Total Return index (^SP500TR) rather than the price-only ^GSPC, which would omit roughly 1.8 to 2 percentage points a year of dividends and overstate the strategy's relative performance.

Each portfolio forms in January of its first year, is equally weighted at inception, and is reviewed every January. Proceeds from a sale are reinvested immediately in the replacement; where no candidate satisfies the buy rule,

proceeds are held in cash and redeployed at the next review. Results exclude transaction costs and taxes. The risk-free rate is 2.5%, the approximate average three-month Treasury bill yield, used in both Sharpe and Sortino.

All performance and risk figures are computed from the realized annual equity curve produced by the simulation. None are estimated.

2.1 One rule was not tested

The strategy as designed carries four sell rules. The third — sell on a year-over-year revenue decline — requires annual fundamentals back to 1995, which the price data source does not provide. It is implemented in the code but disabled. Everything reported here reflects a three-rule strategy, and the fourth rule's contribution is unknown.

3. Portfolio Construction

Hold the ten largest S&P 500 companies by market capitalization, reviewed each January. A position is sold if any of the following holds:

Rule	Trigger	Firings (1995–25 / 2000–25)
1. Rank exit	No longer in the top ten by market capitalization	54 / 45
2. Drawdown	25% or more below the trailing 52-week high	15 / 14
3. Revenue decline	Revenue contracted year over year	Not tested — Section 2.1
4. Stagnation	Held ten or more years at a position CAGR below 8%	2 / 3

A sold position is replaced by the largest company not already held, subject to that company having posted a positive return in the prior calendar year, and excluding any name sold earlier in the same review. Weights are left to drift after formation; a position is trimmed back to equal weight only if it exceeds 25% of portfolio value.

4. Results

$10,000 initial investment, no transaction costs, no taxes.

4.1 1995–2025 (31 years)

Metric	Strategy	S&P 500 TR	Verdict
CAGR	11.03%	11.14%	Behind
Final value	$256,287	$264,464	Behind
Arithmetic mean	13.39%	12.76%	Ahead
Volatility	22.41%	18.06%	Worse
Sharpe ratio	0.49	0.57	Worse
Sortino ratio	0.98	1.05	Worse
Maximum drawdown	−45.55%	−37.61%	Worse

Metric	Strategy	S&P 500 TR	Verdict
Calmar ratio	0.24	0.30	Worse
Positive years	74.19%	80.65%	Worse

4.2 2000–2025 (26 years)

Metric	Strategy	S&P 500 TR	Verdict
CAGR	5.98%	8.11%	Behind
Final value	$45,296	$76,022	Behind
Arithmetic mean	7.90%	9.72%	Behind
Volatility	19.87%	17.93%	Worse
Sharpe ratio	0.27	0.40	Worse
Sortino ratio	0.45	0.68	Worse
Maximum drawdown	−44.34%	−37.01%	Worse
Calmar ratio	0.13	0.22	Worse
Positive years	73.08%	76.92%	Worse

The hypothesis is rejected in both windows. The robustness matters: a result that held in one period and reversed in the other would say more about the sample than the strategy. This one does not reverse.

5. Where the Return Went

The 1995–2025 window poses the sharper question. The strategy earned a higher average annual return than the index — 13.39% against 12.76% and ended with less money. Compound return falls below the arithmetic mean by approximately half the variance, and at these volatilities that penalty is 2.51 percentage points for the strategy against 1.63 for the index:

	Strategy	S&P 500 TR
Arithmetic mean	13.39%	12.76%
Less variance drag ($\sigma^2/2$)	−2.51%	−1.63%
Implied compound return	10.88%	11.13%
Actual CAGR	11.03%	11.14%

The approximation lands within 0.15 points of the realized figures. The strategy earned 0.63 points a year more than the index and surrendered 0.88 points more to its own dispersion. The excess return was real; it simply was not large enough to pay for the volatility used to generate it.

This is the practical content of the result. A strategy can be right about direction and still lose, if being right requires carrying enough variance that compounding claws the advantage back. Raw average return is not what an investor keeps.

5.1 The drawdown rule is pro-cyclical

The rule fires after a decline have happened, so it sells into weakness. Whether that is protective or destructive depends entirely on what follows, and the rule cannot know the moment it fires.

In January 2001 it sold Microsoft, Cisco, General Electric and Intel ahead of the dot-com collapse — the exits the strategy was designed to make. In January 2023 it sold Microsoft, Alphabet, Amazon and Apple in a single review, immediately before the year those names drove the market. In January 2009 three positions went to cash because no candidate passed the positive prior-year filter at the bottom of the financial crisis; they were redeployed a year later, having missed the recovery.

A rule triggered by price decline alone has no way to separate a cyclical drawdown from a structural one. Over both windows the cost of the misses exceeded the benefit of the saves.

5.2 Rank exit dominates turnover

Rank exits account for 54 of 79 firings in the longer window and 45 of 67 in the shorter — the mechanical consequence of holding exactly ten names drawn from a pool of twenty. Much of this is noise rather than signal: a company at rank 10 and one at rank 11 differ negligibly, yet the rule treats the boundary as decisive. A buffered version, selling only on a fall past rank 12 or 13, would cut turnover substantially and is the first variant worth testing.

6. Limitations

6.1 Two windows are not a distribution

Both windows end in 2025 and overlap heavily, so they are not independent tests. Thirty-one annual observations with 79 rule firings are a small effective sample, and a handful of events drive most of the outcome. Agreement between the two is reassuring but falls well short of statistical significance.

6.2 One rule untested

As noted in Section 2.1, the revenue-decline rule was not backtested. A fundamentals-based exit might separate structural from cyclical declines where the price-based rule cannot, precisely the weakness identified in Section 5.1. This is the most important open question the test leaves.

6.3 Unmodeled costs

Transaction costs, bid ask spread and taxes are excluded. Turnover averages under three trades a year, so trading costs are immaterial. Taxes are not: on a taxable account the gains realized at each January review would reduce the effective return further, widening the gap to a buy-and-hold index position rather than narrowing it.

6.4 Index membership and price history

The universe is S&P 500 constituents, so a large non-index company would be excluded. Twelve tickers have price histories beginning after 1995, mostly companies that had not yet listed, and the script reports them rather than failing silently; they appear in the rankings only in years after their data begins, so none were dropped from the portfolio they belonged in.

6.5 Annual resolution

The equity curve is annual, so reported drawdowns are measured on year-end values. True intra-year peak-to-trough losses were deeper than those shown for both the strategy and the index.

7. Conclusion

Across both windows tested, the rank signal did not produce excess risk-adjusted return. Over 1995–2025 the strategy matched the index on raw return while taking roughly a quarter more volatility to do it; over 2000–2025 it lost on raw return as well. It underperformed on Sharpe, Sortino, Calmar, maximum drawdown and proportion of positive years in both periods.

The result does not establish that market-capitalization rank is uninformative. It establishes that this implementation, on this data, with one of its four rules untested, does not extract value from it — and that the specific failure is identifiable rather than general: an exit triggered by price alone cannot distinguish a cyclical drawdown from a structural one, and a portfolio of ten names carries enough variance that a modest edge in average return does not survive compounding.

7.1 What to test next

- Buffer the rank-exit rule: sell only on a fall past rank 12 or 13 rather than at the top ten boundary. This rule drives roughly two-thirds of all turnovers, much of its likely noise.
- Vary the drawdown threshold. 25% was chosen without optimization; a sensitivity analysis across 15% to 40% would show whether the result is robust or an artifact of one number.
- Source annual fundamentals and enable the revenue-decline rule, then retest. Most likely candidate to address the Section 5.1 failure.
- Widen the portfolio to 15 or 20 names and retest. Section 5 identifies variance, not selection, as the binding constraint in the longer window; more positions would reduce dispersion directly.
- Extend to 1990, the rankings already support it, and run a walk-forward split, fitting on the earlier period and testing on data not examined while designing the rules.

8. Reproduction

The full backtest is one Python file with the rankings embedded, requiring only yfinance, pandas and numpy. It runs every window in a single configuration list from one price download and writes the annual equity curve and a complete trade log, with the triggering rule recorded for every firing, per period. Every figure in this note can be regenerated by running it.

Repository: https://github.com/georgebesch123-spec/Mega-Cap-Momentum.git